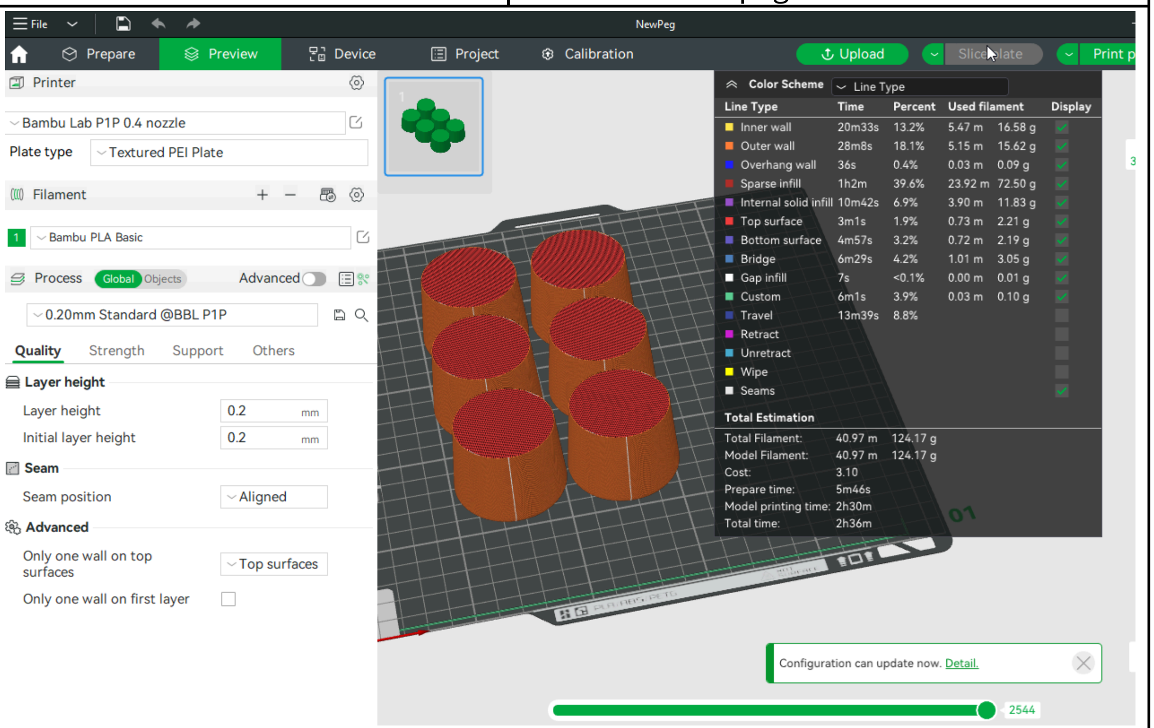
MyDesign Portfolio

Element G

Prototype Construction of Design #1

(make as many copies, updating the design number, as needed)

Date: 4/4-15/25

<table>
<tr><td></td><td>Details</td><td>Tools and Manufacturing Methods used</td></tr>
<tr><td>Step 1</td><td>We created our 2 inch PLA pegs via Onshape</td><td>Onshape</td></tr>
<tr><td>Step 2</td><td>Then we 3d printed them out</td><td>Onshape, Bamboo Studio
Print time for 6 pegs = 2 hours 36 min</td></tr>
<tr><td></td><td colspan="2">
</td></tr>
<tr><td>Step 3</td><td>We then applied the Self-adhesive No Slip Pads onto the bottom of our PLA pegs</td><td>No Slip Pads, Scissors
Yelanon Non Slip Furniture Pads -12pcs 4" Furniture Grippers Hardwood Floors, Non Skid for Furniture Legs,Self Adhesive Rubber Feet, Anti Slide Furniture Floors Protectors for Keep Couch Stoppers [Amazon]</td></tr>
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Step 4	We then marked up the PLA pegs onto the Plexiglass	Dry Erase Marker, PlexiGlass Board. Tape Measure
Step 5	We then used super glue to glue the PLA Pegs onto the Plexiglass	Gorilla Super Glue, PLA Pegs, PlexiGlass Board
Step 6	We then cut the rubber panel connector	handsaw.
Step 7	We then connect the rubber panel connector to our Plexiglass	Plexiglass, Rubber Panel Connector
Step 8	We repeat the process 7 times with 3 boards having custom configurations of the pegs	Gorilla Super Glue, PLA Pegs, PlexiGlass Board

Notes of the progression from your blueprint to your (various) physical prototype(s), including why and how aspects of your design may have changed:

Final Prototype Construction of Design #1 *(change this number to match the design number you're on)*

Date: 4/20-30/25

	Details	Tools and Manufacturing Methods used
Step 1	We started converting the real life changes onto our Onshape models	Onshape
Step 2	We added the extra Pegs that were needed for stabilizing the structure	Onshape
Step 3	Then we created the Plastic Panel connectors that were used to attach each panel together	Onshape
Step 4	We then realized that the proportions for the plastic Panel Connectors weren't right to attach to the actual Panels in Onshape you we had to completely start it over from scratch	Onshape
Step 5	AFter completing the Plastic Panel Connector models in Onshape, we	Onshape

	used mate connectors to attach the to the assemble	
Step 6	The we had to create a model for a custom, smaller cable trough in Onshape	Onshape
Step 7	We attached the Pegs to a smaller Board model and then cut the Plastic Panel Connector and attached it to that as well.	Onshape
Step 8		

Material name	**Description**	**Where material was purchased**	**Cost per unit**	**# units**	**Total cost per item**
Plexiglass	Less fragile/flexibile/ sturdy glass	Acquired from the storage room	0	8-9	0
3D filament	3D printed pegs	Amazon	$0.56	81	$45.36
Yelanon Non Slip Furniture Pads -12pcs 4"	No slip pads	Amazon	$0.6741	12	$8.09
Flat Plexiglass Panel Connectors		Amazon			$19.95
Gorilla Super Glue		Amazon			$8.69
Hot Glue Gun					$3.87

Detailed sketches, which might include CAD if you have learned it in your class:

Design Requirement	How will this prototype allow you to collect objective, measurable data?

Must be with the measurement constraints; Length -236, Width-18, Height-3	It will allows us to see how our final product will fit into the given space and how will will inevitably have to adjust and create new arranges for the future cable troughs
The design must be strong enough to support the weight of a standing and mobile average person	This will allow us to collect data on how strong the cable trough has to be within compromising the design of lightweightness of the final product

Attribute	**Description**	**Describe your prototype's functionality to test this attribute**
Strength	The Cable Trough has to be strong enough to support people walking and standing on it	Our Cable Trough is made of very strong, high quality acrylic that is supported by PLA Pegs which improves the spread of the force from the weight of the person standing on it
Weight	The product has to be lightweight and easy to carry	Since we used Acrylic or our cable trough, we had made it so that our Cable Trough will be not very heavy but still maintain it's strength

Portability	The product can't be a weird shape that'll be hard to store, it has to be easy to put away anywhere	Our Cable Trough design allows must Panels to be stacked upon each other easily with friction pads on the bottom of each peg to ensure stability